SHIV NADAR UNIVERSITY

S.No. SNU/ 197954

SHIV NADAR UNIVERSITY

TEST ANSWER BOOK

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Course Code ECO 354 Course Title Public Economics

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Marks	20	23												43/50
Q.No.	13.	14.	15.	16.	17.	18.	19.	20.	21.	22.	23.	24.	Sub Total	
Marks														

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ECO 354 → Midsem Exam

A1: Two oil refineries

Market price = \$3/gallon (Arbitrary choice of currency)

Cost of production = \$2/gallon

Production = y_i

$i \in \{1, 2\}$

$S_1 = 0.01\, y_1^2$

$S_2 = 0.01\, \frac{y_2^2}{2}$

Capacity = 200 gallons

a) NO REGULATIONS

$\max_{y_i} \quad 3y_i - 2y_i$ (Profit maximisation)

$$\Rightarrow \boxed{\begin{array}{l} \max_{y_i} \; y_i \\ \text{s.t } y_i \leq 200 \end{array}}$$ // This is the optimisation problem

Clearly,

Optimal production at corner pt

$\Rightarrow y_1 = \underline{200}$ gallons

$y_2 = \underline{200}$ gallons

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Smog:

- $S_1 = 0.01\, y_1^2$
  $= 0.01 (200)^2 = \frac{1}{100} \times 40000 = 400$ units
- $S_2 = 0.01\, y_2^2/2 = \frac{1}{200} \times 40000 = 200$ units

$\Rightarrow$ TOTAL SMOG $= 400 + 200 = \underline{600 \text{ units}}$

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b) INTERNALISING THE EXTERNALITIES

For firm 1,

$$\max_{y_1} \quad y_1 - s_1$$ // Including damage f^n as cost

$$\Rightarrow \boxed{\begin{array}{l} \max_{y_1} \quad y_1 - 0.01 y_1^2 \\ \text{s.t.} \quad y_1 \leq 200 \end{array}}$$

- First order condition:

$$1 - 0.01(2y_1) = 0$$

$$\therefore 1 = \frac{2y_1}{100}$$

$$\Rightarrow \underline{y_1 = 50 \text{ gallons}}$$

Similarly,
For firm 2,

$$\max_{y_2} \quad y_2 - s_2$$

$$\Rightarrow \boxed{\begin{array}{l} \max_{y_2} \quad y_2 - 0.01 y_2^2/2 \\ \text{s.t.} \quad y_2 \leq 200 \end{array}}$$

- First order condition:

$$1 - \frac{0.01}{\cancel{2}}(\cancel{2}y_2) = 0$$

$$\therefore 1 = 0.01 y_2$$

$$\Rightarrow \underline{y_2 = 100 \text{ gallons}}$$

★ These are the socially optimal levels of production.

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A2:

Smog:

$$d_1 = 0.01 y_1^2$$
$$= 0.01 (50)^2$$
$$= \frac{1}{100} \times 2500$$
$$= \underline{25 \text{ gallon units}}$$

And,

$$d_2 = 0.01 y_2^2/2$$
$$= \frac{1}{100} \times \frac{10000}{2}$$
$$= \underline{50 \text{ units}}$$

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$\therefore$ TOTAL SMOG $= 25 + 50 = \underline{75 \text{ units}}$
(Socially efficient level)

c) The socially optimal production for both firms is way lesser than the private market outcome for both firms.
This happens because smog has a damage function associated with it and is hence, clearly a negative externality. We know that goods with negative externalities are overproduced in the market
However, since both firms have the same capacity and profit f^n, they produce the same amount of oil in the case with no regulations. But when the externalities are internalised, firm 1 produces only half as much as firm 2 (plant 2), because it pollutes double as much as firm 2.

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A2: Sebastian and Mia

$U_i(x_i, S) = \ln(x_i - 3) + \ln(S-6)$

$\omega_s = 36$

$\omega_m = 24$

$S = s_s + s_m$

$P_{x_i} = P_s = 1$

a) For Sebastian

$$U_s(x_s, S) = \ln(x_s - 3) + \ln(S-6)$$
$$= \ln(36 - s_s - 3) + \ln(s_s + s_m - 6)$$
$$= \ln(33 - s_s) + \ln(s_s + s_m - 6)$$

To maximize

$$\max_{s_s} \ln(33 - s_s) + \ln(s_s + s_m - 6)$$

First Order Condition:

$$\frac{\partial U_s}{ds_s} = \frac{-1}{33 - s_s} + \frac{1}{s_s + s_m - 6} = 0$$

$$\Rightarrow \frac{1}{s_s + s_m - 6} = \frac{1}{33 - s_s}$$

$$\Rightarrow s_s + s_m - 6 = 33 - s_s$$

$$2s_s = 39 - s_m$$

$$s_s = \frac{39 - s_m}{2} \quad \text{--- (1)}$$

For Mia,

$$U_m(x_m, S) = \ln(x_m - 3) + \ln(S-6)$$
$$= \ln(24 - s_m - 3) + \ln(s_s + s_m - 6)$$
$$= \ln(21 - s_m) + \ln(s_s + s_m - 6)$$

P.T.O.

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To maximise

$$\max_{s_m} \ \ln(21 - s_m) + \ln(s_s + s_m - 6)$$

First order condition:

$$\frac{dU_m}{ds_m} = \frac{-1}{21 - s_m} + \frac{1}{s_s + s_m - 6} = 0$$

$$\therefore \frac{1}{21 - s_m} = \frac{1}{s_s + s_m - 6}$$

$$\Rightarrow 21 - s_m = s_s + s_m - 6$$

$$\therefore 2s_m = 21 + 6 - s_s$$

$$\Rightarrow 2s_m = 27 - s_s$$

$$\therefore s_m = \frac{27 - s_s}{2} \quad \text{——— (II)}$$

Sub (I) in (II)

$$s_m = \frac{1}{2}\left[27 - \frac{39 - s_m}{2}\right]$$

$$= \frac{1}{2}\left[\frac{54 - 39 + s_m}{2}\right]$$

$$= \frac{15 + s_m}{4}$$

$$\therefore 4s_m = \frac{15}{4} + s_m$$

$$3s_m = 15$$

$$\Rightarrow \underline{s_m = 5}$$

Sub in (I),

$$s_s = \frac{39 - 5}{2} = \frac{34}{2} = \underline{17}$$

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Hence,

Seats paid for by Sebastian = 17

Seats paid for by Mia = 5

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⇒ TOTAL SEATS BUILT IF NO INTERVENTION = 17 + 5
= 22

BONUS ATTEMPT → Clearly, we see that both s_s and s_m depend upon the incomes of Sebastian and Mia respectively.

So if income ↑, no. of seats paid for ↑

We also see that the no. of seats one person pays for varies negatively with the contribution of the other person.

∴ If income of 2 ↑, no. of seats paid for by 1 ↓

From these two reasons, it is evident that the person with the higher income will pay for more seats and vice versa.

⇒ & the equilibrium DOES indeed depend on the income distribution.

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b) To get the socially optimal number of seats,

Let us think of Sebastian and Mia as one person.

The combined income is 36 + 24 = 60

The utility f^n is

For socially optimal no. of seats, $MRS^S + MRS^M = MRT$

$$u(x, S) = \ln(x-3) + \ln(S-6)$$

$$\therefore u(x, S) = \ln(60-S-3) + \ln(S-6)$$

$$= \ln(57-3) + \ln(S-6)$$

$MRS^S = \dfrac{x_s - 3}{S-6}$ $\quad MRS^M = \dfrac{x_m - 3}{S-6}$ $\quad MRT = 1$

$$\max_{S} \; \ln(57-S) + \ln(S-6)$$

First order condition:

$$\frac{-1}{57-S} + \frac{1}{S-6} = 0$$

$$\therefore 57 - S = S - 6$$

$$\therefore 2S = 63$$

$$\Rightarrow S = \frac{63}{2} = \underline{31.5}$$

Since 31.5 is not possible,
and we observe that the utility gained from building 31 and 32 seats is the same.
So for convenience,
let us say,
socially optimal number of seats = 31 ✗

→ This differs from the answer in case a) because people do not contribute as much as they should and this is called the PROBLEM OF COMMONS.
They are only maximising their own utility and not looking at the society as a whole.

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c) For Sebastian,

(6 seats are added)

$$\max_{S_s} \ln(30 - S_s) + \ln(S_m + S_s - 6)^{+6}$$ // Real income has gone down

FOC: $$\frac{1}{30 - S_s} = \frac{1}{S_m + S_s - 6 + 6}$$

$$\therefore 30 - S_s = S_m + S_s - 6 + 6$$

$$\therefore 2S_s = 36 - S_m \qquad 2S_s = 30 - S_m$$

$$\Rightarrow S_s = 18 - \frac{S_m}{2} \quad \text{—— (1)}$$

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For Mia,

$$\max_{s_m} \ln(18 - s_m) + \ln(s_m + s_s - 6) + 6$$

FOC: $\frac{1}{18 - s_m} = \frac{1}{s_m + s_s - 6 + 6}$

$\Rightarrow 18 - s_m = s_m + s_s - 6 + 6$

$\therefore 2s_m = 24 - s_s$ $\qquad 2s_m = 18 - s_s$

$\therefore s_m = 12 - \frac{s_s}{2}$ ——— (II)

Sub (I) in (II)

$$s_m = 12 - \frac{1}{2}\left[18 - \frac{s_m}{2}\right]$$

$$= 12 - 9 + \frac{s_m}{4}$$

$$\frac{3s_m}{4} = 3$$

$\Rightarrow s_m = 4$ seats $\qquad s_m = 2$

Sub in (I)

$$s_s = 18 - \frac{4}{2}$$

6 $\Rightarrow s_s = 16$ seats $\qquad s_s = 14$

$\therefore$ TOTAL SEATS $= 4 + 16 + 6$

$= 26$ seats $\qquad 22$.

$\Rightarrow$ The no. of seats has INCREASED as compared to part (a) ✓ $\qquad$ remains same as part (a).

$\Rightarrow$ Social optimum NOT achieved. ✓

Both Sebastian and Mia contribute lesser as their real income has gone down and their utility maximisation point has changed.

d) For Sebastian

$$\max_{s_s} \ln(33 - s_s) + \ln(s + 6 - 6)$$

$$\Rightarrow \max_{s_s} \ln(33 - s_s) + \ln(s)$$

FOC →

$$\frac{1}{33 - s_s} = \frac{1}{s_s + s_m}$$

$$\Rightarrow 33 - s_s = s_s + s_m$$

$$\therefore 2s_s = 33 - s_m$$

$$s_s = \frac{33 - s_m}{2} \quad \text{—— (I)}$$

For Mia

$$\max_{s_m} \ln(21 - s_m) + \ln(s_s + s_m)$$

FOC →

$$\frac{1}{21 - s_m} = \frac{1}{s_s + s_m}$$

$$\Rightarrow 21 - s_m = s_s + s_m$$

$$\Rightarrow 2s_m = 21 - s_s$$

$$\therefore s_m = \frac{21 - s_s}{2} \quad \text{—— (II)}$$

Sub (I) in (II)

$$s_m = \frac{1}{2}\left[21 - \frac{33 - s_m}{2}\right]$$

$$= \frac{1}{2}\left[\frac{42 - 33 + s_m}{2}\right] = \frac{1}{4}(9 + s_m)$$

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$\therefore \quad 4s_m = 9 + s_m$

$3s_m = 9$

$\Rightarrow \underline{\underline{s_m = 3}}$ seats

Sub in (1)

$s_s = \dfrac{33 - s_m}{2}$

$= \dfrac{33-3}{2} = \dfrac{30}{2} = \underline{\underline{15 \text{ seats}}}$

$\therefore$ TOTAL NO. OF SEATS $= 6 + 3 + 15$

$= \underline{\underline{24}}$

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$\Rightarrow$ The no. of seats has reduced from the level in (c)

$\Rightarrow$ Here also, we see that both Sebastian and Mia reduce their contribution and hence we are closer to the social optimum than in (a) but nowhere near it.

Since both see that quite a few seats have already been paid for, they decide to pull some money out and increase private consumption instead.

e) The government can reduce the cost of building a venue seat.

Let's say the new price $= p$, $p < 1$

For Sebastian,

$\max_{s_s} \; \ln(33 - ps_s) + \ln(s_s + s_m - 6)$

FOC $\rightarrow \dfrac{1}{33 - ps_s} = \dfrac{1}{s_s + s_m - 6}$

$\therefore 33 - ps_s = s_s + s_m - 6$

$\therefore (1+p)s_s = 39 - s_m$

$\therefore s_s = \dfrac{39 - s_m}{1+p}$ — (I)

Similarly,
For Mia

$\max_{s_m} \ln(21 - ps_m) + \ln(s_s + s_m - 6)$

FOC $\rightarrow \dfrac{1}{21 - ps_m} = \dfrac{1}{s_s + s_m - 6}$

$\therefore 21 - ps_m = s_s + s_m - 6$

$\therefore (1+p)s_m = 27 - s_s$

$\therefore s_m = \dfrac{27 - s_s}{1+p}$ — (II)

(I) + (II)

$s_s + s_m = \dfrac{39 + 27 - s_m - s_s}{1+p} = 31$

$\Rightarrow \dfrac{39 + 27 - 31}{1+p} = 31$

$\dfrac{35}{1+p} = 31$

$1 + p = \dfrac{35}{31}$

$\Rightarrow p = \dfrac{35}{31} - 1$

$\Rightarrow p = \dfrac{4}{31}$ units

Govt provide 30 seats and tax Seb & Mia accordingly.
Seb and Mia pay 3/5th and 2/5th shares respectively.
there will be no govt provision and no of seats will equal the social optimum.

(3)

$\Rightarrow$ This is the price the government should set to get socially optimal amount of seat provision.

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